Canada Border Services Agency | Agence des services frontaliers du Canada

Peer Review

Remote Reporting Program

Client Reporting and Engagement System (CRES)/ReportIn

Executive Summary

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CRES Peer Review

CBSA Background

The Canada Border Services Agency (CBSA) is committed to using technology responsibly. The Agency developed a Remote Reporting Program consisting of a CBSA-facing technical backend system, Client Reporting and Engagement System (CRES), and a public-facing mobile device/smartphone application (app), ReportIn, where individuals subject to reporting conditions are able to report remotely to the CBSA by collecting facial biometrics as a form of authenticating the individual's identity.

This peer review is required by the Treasury Board Secretariat (TBS) to ensure compliance with its policy direction based on the TBS Directive on Automated Decision-Making on responsible use of artificial intelligence in government. This peer review was completed by the Data Science Division, Chief Data Officer Branch, at Employment and Social Development Canada.

The goal of the peer review, by validating the CBSA's Algorithmic Impact Assessment, the quality of the system, and the quality assurance and risk mitigation measures, was to provide an assessment on the responsible use of the system and residual risk identification of operating the system.

The Remote Reporting Model

This peer review is related to the development and deployment of a self-reporting app (called ReportIn) made by CBSA for CRES (Client Reporting & Engagement System). This is part of ongoing improvement and expansion of options to Alternatives to Detention (ATD).

The ReportIn app lets an individual do self-reporting with their smartphone or mobile device without having to call or report in-person to a CBSA officer. This self-reporting process uses facial photos as biometric markers to identify the reporting individual. This should remove pressure on CBSA officers and reporting individuals.

Eligible individuals for enrollment must be at least 18 years of age, and they must have a smartphone or mobile device (with a camera and a GPS), a telephone number, and a valid email address.

When an individual is enrolled in the remote reporting program, they install the app on their smartphone or mobile device. Five baseline pictures of their face (front, left side, right side, inclined up and declined down) are registered in their file. At that moment, the CBSA officer also decides or inputs the frequency of reporting in CRES. At the scheduled times, the app informs the user that they should self-report. When this happens, the individual needs to take five pictures with the same registered poses in an indicated order (selected randomly). The images are compared to the baseline in CRES. If the biometric identification succeeds, the reporting is a success. Otherwise, the reporting individual can be asked to retake their pictures, call-in, or report in-person. In the case of significant changes in appearance (ex.: a major change of haircut, transitioning, weight loss/gain), the ATD Monitoring Centre officer can ask the reporting individual to retake their baseline pictures using the app. When the pictures are sent to CRES, the app also sends the location of the individual.

During the Client Case Review process, cases can be sent to Regional Review under certain circumstances, such as when the individual fails to report or comply with ATD Monitoring Centre Officer's requests. A case is considered concluded when it meets certain criteria, including completion of the Regional Review, moving the individual's status to inactive, or successful reporting by the individual.

When the app takes facial pictures of the reporting individual, the face detection, and its pose estimation (to confirm that there is indeed a person in front of the camera) is performed by GoogleML toolkit. The face identification is performed by the Face Comparison module of Amazon Rekognition.

In the first phase of deployment of the app, no biometric decision will be made without a human confirming (or refuting) the results. This means that there is no danger in this phase of the project for any clients.

Peer review by ESDC

Methodology

The review focus was on the app operation and on the face identification process.

For the self-reporting process, we verified that there were no logistical pitfalls of any kind in the enrollment process, the self-reporting process itself, and the consequences of the self-identification process (i.e., what happens if the biometric identification is a success or a failure).

For the face identification process, we verified if the selected system was adequate for the needs of the app, that it was thoroughly tested, and that it was unbiased.

Major Issues

There are no major issues identified in this project.

Minor issues

The only potential issue identified is the fact that the Amazon "Rekognition" Face Comparison module was not formally tested by the CRES development team. In absence of this formal test, the only sources of information available about this face identification system are academic publications on the bias of most of the Amazon "Rekognition" modules.

The reason why this only a minor issue is because the CRES development team have wisely chosen to first deploy this system with a complete human overview instead of directly trusting the automatic decisions of the system. Furthermore, if the CRES development team follows our recommendations, we believe that this issue could be mitigated.

ESDC Recommandations & CBSA Response

ESDC Recommandation

Our recommendations are twofold.

1. A formal test dataset should be created for the purpose of evaluating bias when using the Amazon Rekognition Face Comparison module. To accomplish this goal, this dataset should include at least 5-10 persons from all possible subgroups (different combination of attributes: age, gender, ethnicity) that would compose the client base. Considering that there are a lot of different subgroups, generating this dataset could be complicated. Indeed, if we estimate that

there are, at least, 5 age groups (18-24, 25-34, 35-49, 50-64, 65+), 3 gender groups (male, female, non-binary/other)[1], and, let's say, 6-7 ethnic groups (at minimum: Caucasian, Arab, Black, Asian, Hispanic, Indigenous)[2], this suggests at minimum 90 subgroups (which would result in a dataset of 450-900 different persons). Since the goal is to test the bias of the CRES *ReportIn* system in a way that is similar to the system use case, two series of the five pictures (with some slight variation in background and lighting if possible – let's say imagine that those pictures were simply taken in another room) for every person in this dataset would be needed.

Simplifying the number of subgroups could be possible by changing the ethnic groups for simply the skin tone groups (light or dark skinned); this would have the problem of being even more arbitrary, but it could still be useful.

A way to bypass the problem of finding or collecting this dataset would be to use the data from the first phase of deployment as a dataset to experiment with and measure the Face Comparison bias. This, of course, could imply some security assessment for the extraction of images from the CRES dataset (unless the analysis is performed directly on the images from the dataset).

2. With a dataset containing two sets of five images of a lot of people with varied traits (age, gender, ethnicity), it is possible to perform statistical analysis using the Face Comparison confidence output to evaluate its bias.

   The first step is to perform face comparison between both set of five pictures for all the persons in the test dataset. This first test should help find a more appropriate threshold for all the subgroups if needed (perhaps 99% confidence can't be achieved for certain subgroups).

   Then, the face comparison should be performed between all people from a same subgroup. What needs to be evaluated is the average confidence difference between invalid pairs inside the same subgroup; this value should be similar for all subgroups. If it's not the case, it represents a bias where the model can't easily differentiate between people of this subgroup. This analysis should be performed on a lot of combined subgroups (young vs old, light skinned vs dark skinned, male vs female vs non-binary) since bias could be easier to detect on these larger subgroups.

   Once we know what (if any) subgroups are biased (and how much), it is possible to adjust the confidence of the predictions for these subgroups. We suggest something akin to Platt scaling to perform this post-hoc output adjustment. Of course, there are other, more complex, solutions to the problem of unbiasing a black box system, but this would be our first recommendation.

[1] We recognize that there are more than 3 gender choices. This an arbitrary division of genders with the goal of limiting the dataset size.

[2] This number of ethnic groups is well under the Canada census number of options; this is just an example.

CBSA Response

Above and beyond meeting the necessary requirement of the Directive on Automated Decision-Making, ReportIn has also taken active steps to identify and mitigate additional consideration and unintended harm by:

- Leveraging existing expertise at the CBSA in both Biometrics, data and Responsible AI;
- Consulted and reviewed existing research into Biometric systems and potential social implications;
- Actively tried to apply a co-design approach with the CBSA's social stakeholder and equity deserving representative to ensure equitable, respectful and representative outputs and application; and
- Aligning with existing Automated Decision-Making and GBA Plus directives by conducting extensive social review, engaging with legal, communications and privacy and co-design with equity deserving partners.

The CBSA is committed to:

1. Having the Agency's Office of Biometrics and Identity Management evaluate the program approximately six months to one year after CRES is launched to assess the following:
    a. Human Evaluation vs Amazon Rekognition results
    b. Whether there is bias in relation to ethnicity, age, gender
2. Having the Agency's Office of Biometrics and Identity Management (OBIM) study the data set approximately six months to one year after launch to conduct a statistical analysis on the possibility of bias using the Face Comparison. The results of this evaluation will assist the CBSA in adjusting CRES as necessary.
3. Continuously reviewing and monitoring complaints received from users of the system to address and document any possibility of bias.